Local Action Diagrams

Super awesome package for local action diagrams.

0.1

29 January 2023

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Chapter 1

RSGraphs

An RSGraph is a graph $\Gamma = (V, A, o, r)$ that consists of a vertex set V, an arc set A, a map $o : A \to V$ which assigns each vertex to an *origin*, and a bijection $r : A \to A$ which assigns each arc to a *reverse* arc. There is also a map $t := o \circ r$ which assigns each arc to a *terminus* vertex. Note the RSGraphs are allowed to contain loops and parallel arcs.

We created the RSGraph object for this package due to our use of graphs having a specialised definition. For example, every arc in an RSGraph must have a reverse arc associated with it while a *Digraph* from the Digraphs allows for arcs that do not have an associated reverse arc. Creating a new object category for RSGraphs allows for both better error checking when using RSGraphs and to avoid conflicting definitions, such as the definition of a cycle for a Digraph and an RSGraph. We provide functionality to convert between an RSGraph and a *Graph* from the GRAPE package or a *Digraph* from the Digraphs package.

The name "RSGraph" arises from Reid and Smith, the authors of the local action diagram paper. This name was chosen to avoid conflicting with the GRAPE package.

1.1 Creating RSGraphs

1.1.1 IsRSGraph (for IsObject)

▷ `IsRSGraph(`*graph*`)` (Category)

Returns: `true` if *graph* is of the category `IsRSGraph` and false otherwise.

Every RSGraph object belongs to the category `IsRSGraph`. Furthermore, every RSGraph is an immutable attribute storing object.

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